CSE-251

Electronic Circuits

Lecture-01

Syed Jamiul Alam
Lecturer
Department of CSE

EAST WEST UNIVERSITY
excellence in education
Department of
Computer Science & Engineering

Today's Overview:

- ***Introduction to the Course***

Credit Hours and Teaching Scheme:

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14^{th} Week	2 Hours/Week for 13 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14^{th} Week

Course Objective:

The subject aims to provide students with an understanding and capability to use the basic electronic abstractions to analysis and design circuits and systems built with electronic circuit elements.

This course provides fundamental knowledge of how complex devices such as semiconductor diodes, operational amplifiers (op-amp), bipolar and field effect transistors are modeled and used in the design and analysis of useful practical circuits. Besides, this course also emphasizes practical implementation of building, testing and performance analysis of electronic circuits.

Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE345 Digital Logic Design, CSE350 Data Communications, CSE360 Computer Architecture, CSE442 Microprocessor and Microcontrollers and CSE490 VLSI Design.

Teaching Materials:

❑ **Textbook:**

1. Electronic Devices and Circuit Theory- Robert L. Boylestad and Louis Nashelsky
2. Electronic Devices and Circuits – David A. Bell
3. And of course, the class lectures! (will save you in time)

❑ **Lab Manual:**

- Lab manual will be provided.

❑ **Project Description:**

- Project description will be provided.

❑ **Equipment/Software:**

- Digital trainer board, and Computer & PSpice Software.

Classroom Etiquette:

- ***Arrive in class on time*** (10 mints max will be tolerated, but conditions applied for every next time)
- ***Avoid giving proxy attendance*** (Fear the GOD, after 100 years, none of us will be here, so what's the point?)
- ***Keep classroom neat and clean*** (You're not a kid anymore!)
- ***Preferably switch the cell phones*** (can be kept in silence though)***, tabs, laptops OFF while you're in the class.***
- ***Avoid distracting conversations*** (Relevant questions are definitely encouraged)
- ***There will be zero tolerance for cheating by any means*** (Well, this is solid seriously meant to you all)

Course Topics, Teaching-Learning Method, and Assessment Scheme

Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of COs	Exam Mark
			C2	C3	C4		
Operation and characteristics of semiconductor diode, Load-line analysis.	Lecture, Slides, Class Discussion, Discussion with Instructor/TA	CO1	3	4		7	Mid Semester Exam (30)
Applications of Diode: Rectifier circuits, Clipper and Clamper, Zener Diode.	Do	CO2		4	5	9	
Device Structure and Physical Operation of BJT, Modes of Operation, Current-Voltage Characteristics, BJT as amplifiers and switch.	Do	CO1	3			3	
BJT circuits at DC, Biasing in BJT amplifier circuits.	Do	CO2			5	5	
Characteristics of the ideal Op-amp, Basic Comparator circuits, Inverting and non-inverting amplifiers, Voltage follower.	Do	CO1	3	3		6	

Course Topics, Teaching-Learning Method, and Assessment Scheme

Applications such as Adder and Difference amplifiers, Integrator and Differentiator, and Instrumentation amplifiers.	Do	CO3		4	9	13	Final Exam (30)
Device Structure and Physical Operation of MOSFET, Modes of Operation, Current-Voltage Characteristics, MOSFET as an Amplifier and Switch.	Do	CO1	3	4		7	
DC biasing and small signal operations of MOSFET, Small signal equivalent models of MOSFET, Logic circuit using n-MOS and p-MOS.	Do	CO2		3	7	10	

Overall Assessment Scheme

Assessment Area	CO				Total	PO Marks	
	CO1	CO2	CO3	CO4		PO1	PO2
Class Test/Quiz	5	5	5		15	5	10
Mid Semester Exam	16	14			30	16	14
Final Exam	7	10	13		30	7	23
Laboratory Performance and Lab Exam				15	15	15	
Mini Project				10	10	10	
Total	**28**	**29**	**18**	**25**	**100**	**53**	**47**

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

QUESTIONS?

See You In The Next Class